FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VI

JEE (Main)-2022

TEST DATE: 13-02-2022

ANSWERS, HINTS & SOLUTIONS

Physics PART – A

SECTION – A

1. A

Sol. The pendulum would strike the xy plane
When the string is cut z = h = 5m and v = 4 m/s parallel to the y-axis
The time taken by the bob to hit the xy plane is $5 = \frac{1}{2} \times 10 \times t^2$
$\therefore$ t = 1 sec
During this time it covers a distance 4 m along the y-direction. The point where it hits the xy plane along the x-axis is 1 m because the radius is 1m. Hence the bob hits the xy plane at (1m, 4m)

2. D

Sol. $R = \frac{\rho_0}{A}\left(x + \frac{1}{2}\alpha x^2\right)$
Also, R = $\rho \frac{x}{A}$
Taking differentiation of (1) and (2), we get
$dR = \frac{\rho_0}{A}(dx + x\alpha\ dx)$ and $dR = \frac{\rho}{A} \cdot dx$
$\therefore \frac{\rho_0}{A}(dx + \alpha x dx) = \frac{\rho}{A} dx$
$\therefore \rho = \rho_0(1 + \alpha x)$

3. B

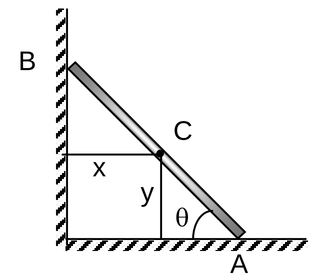
Sol. At any instant of time the rod makes an angle θ with horizontal, the x and y coordinates of centre of rod are

$x = \ell\cos\theta \qquad y = \ell\sin\theta$

$\therefore x^2 + y^2 = \ell^2$

Hence, the centre C moves along a circle of radius with centre at O.

$\therefore$ velocity vector of C is always directed along the tangent drawn at C to the circle of radius ℓ whose centre lies at O.



4. B

Sol. P = VI

For secondary :

$$V_2 = \frac{P_2}{I_1} = \frac{500}{12.5} = 40 \text{ volts}$$

For an ideal transformer (100% efficient)

$P_{input} = P_{output}$

$\Rightarrow V_1I_1 = V_2I_2$

$$\Rightarrow I_1 = \frac{V_2I_2}{V_1} = \frac{40(12.5)}{40\times5} = 2.5\text{ A} \qquad \left[\because \frac{n_1}{n_2} = \frac{V_1}{V_2} \Rightarrow \frac{5}{1} = \frac{V_1}{40}\right]$$

5. D

Sol. Let the value of 'a' be increased from zero. As long as a $\le$ ug, there shall be no relative motion between m_1 or m_2 and platform, that is m_1 and m_2 shall move with acceleration a. As a > μg the acceleration of m_1 and m_2 shall become ug each.

Hence at all instants the velocity of m_1 and m_2 shall be same

$\therefore$ The spring shall always remain in natural length.

6. C

Sol. At any time

$$F_{thrust} - F_{friction} = ma'$$

$$\rho av^2 - \mu mg = ma'$$

$$\rho a(2gh) - \frac{a}{A}(\rho Ah)g = (\rho Ah)a'$$

$$2\rho agh - \rho agh = \rho Aha' \,;\; \frac{a}{A}g = a'$$

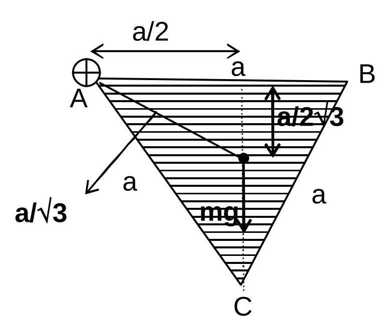
7. C

Sol. Torque about A

$$mg\frac{a}{2} = I\alpha$$

$$\Rightarrow \alpha = \frac{mga}{2I}$$

$$\Rightarrow \text{acceleration } \frac{a}{\sqrt{3}}\alpha = \frac{mga^2}{2\sqrt{3}I}$$



8. B

Sol. Since spherical shell b is earthed therefore potential will be zero.

i.e., $\frac{KQ}{b}+\frac{KQ'}{b}=0 \Rightarrow Q'=-Q$

where Q is total charge inside A and Q′ is charge on sphericla shell B
Therefore field outside B will be zero.
And electric field inside A will be

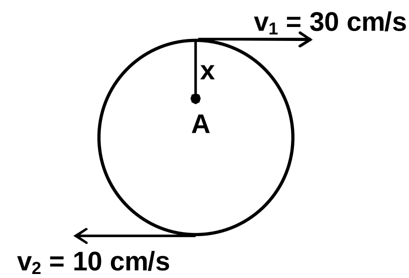
$E=\frac{KQ}{a^3}r$ (linearly increasing)

Between spherical shell A and B

$E=\frac{KQ}{r^2}$

9. D

Sol. Let x be the distance of axis of rotation from the top of sphere A (downward).
Since angular velocity of the top and bottom point of sphere A will be same, therefore,



$\omega=\frac{30}{x}=\frac{10}{(2R-x)}$

$\frac{30}{x}=\frac{10}{(20-x)}$

$\Rightarrow x = 15$

$\therefore$ Distance from ground
= 2R + (2R – x) = 20 + 5 = 25 cm

10. C

Sol. $f_1 = f_2$

$\lambda_1 < \lambda_2$

$c = f\lambda$

$c_1 < c_2$

$c=\sqrt{\frac{T}{\mu}}$

$\mu_1 > \mu_2$

11. C

Sol. At the surface of current carrying wire,

$B=\frac{1}{2}\mu_0 jr$

$j \propto \frac{B}{r}$

As slope of wire P > slope of wire Q

12. D

Sol. $\frac{\mu_2}{v}-\frac{\mu_1}{u}=\frac{\mu_2-\mu_1}{R}$

$\frac{1}{v}-\frac{1.5}{(-u)}=\frac{1-1.5}{-R}$

For v to be positive

$$\frac{1}{2R}-\frac{3}{2u}>0 \Rightarrow u>3R$$

13. A

Sol. N_b reaches maximum when activity of A will be equal to activity of B.

i.e., $\lambda_a N_a = \lambda_b N_b \Rightarrow N_b = \frac{\lambda_a N_a}{\lambda_b}$

14. B

Sol. From COE

$$\frac{1}{2}Kx_0^2+\frac{1}{2}m\left(\sqrt{\frac{3K}{m}}x_0\right)^2=\frac{1}{2}Kx^2$$

$$\frac{1}{2}Kx_0^2+\frac{3}{2}Kx_0^2=\frac{1}{2}Kx^2;\quad 4x_0^2=x^2$$

$x = 2x_0$ (Amplitude)

Equation $y=2x_0\sin\left(\sqrt{\frac{K}{m}}t+\frac{\pi}{6}\right)$

Satisfies condition given in the question.

15. B

Sol. $e = Bv\ell$

$\therefore \frac{de}{dt}=Bv\frac{d\ell}{dt}$ at any instant of time ‘t’

$=Bv\frac{d}{dt}\sqrt{(x^2+y^2}$ where x = OQ and y = OP

$$=Bv\frac{\left(2x\frac{dx}{dt}+2y\frac{dy}{dt}\right)}{2\sqrt{x^2+y^2}}=Bv\frac{\left(x\frac{dx}{dt}+y\frac{dy}{dt}\right)}{\sqrt{x^2+y^2}}$$

$$= Bv\left(\frac{dx}{dt}\frac{x}{\sqrt{x^2+y^2}}+\frac{dy}{dt}\frac{y}{\sqrt{x^2+y^2}}\right)$$

$$= Bv\left(\frac{dx}{dt}\cos\theta+\frac{dy}{dt}\sin\theta\right)$$

$= Bv(v\cos\theta\cos\theta + v\sin\theta\sin\theta)$ [$\because v\cos\theta = \frac{dx}{dt}$, $v\sin\theta = \frac{dy}{dt}$]

$= Bv^2(\cos^2\theta + \sin^2\theta) = Bv^2$

16. B

Sol. no. of photoelectron emitted per second, $n=\frac{\text{Power(watt)}\times\text{Emission\%}}{\text{Energy of 1 photon (in J)}\times 100}$

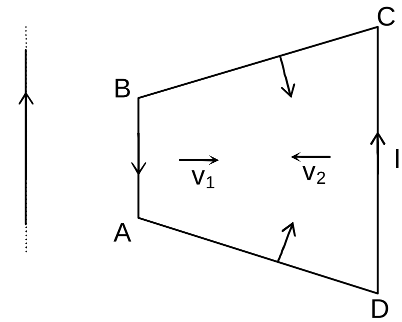
$$=\frac{1.5\times10^{-3}\times0.1}{\frac{1240(\text{nm})(\text{eV})}{400(\text{nm})}\times e\times100}=\frac{0.48}{e}\times10^{-6}$$ (energy of 1 photon = $\frac{1240\text{ nm eV}}{400\text{ nm}}\times e$ joule)

$\therefore$ Photo current = ne = 0.48 μA

17. B

Sol. Magnitude of force on AB (F_1) and magnitude of force on CD (F_2) will be equal in magnitude and opposite in direction.

Hence, $\vec{F}_1 + \vec{F}_2 = 0$. Resultant of $\vec{F}_3 + \vec{F}_4$ will be away from the long straight wire.



18. A

Sol. $-mv_0 + 0 = mv_1 + (2mv_2)$

$-v_0 = v_1 + 2v_2$ …(i)

$e = \frac{v_2 - v_1}{-v_0} = 1$

$v_2 - v_1 = -v_0$ …(ii)

From (i) and (ii)

$v_2 = -\frac{2v_0}{3}$; $v_1 = \frac{v_0}{3}$

$T = \frac{\text{Total distance}}{\text{Velocity of separation}} = \frac{2\pi R}{v_0}$

$\therefore$ Angle rotated by $A = \frac{2\pi}{3}$ radian and angle rotated by $B = \frac{4\pi}{3}$ radian.

19. A

Sol. Velocity of image of particle B

$\vec{v}_B = 5(\hat{i}) + 4(-\hat{i}) = \hat{i}$

Velocity of image of particle A

$v_A = 10(-\hat{i}) + 4(-\hat{i}) + 10\hat{j} = \left[-14(\hat{i}) + 10\hat{j}\right]$

Relative velocity of image

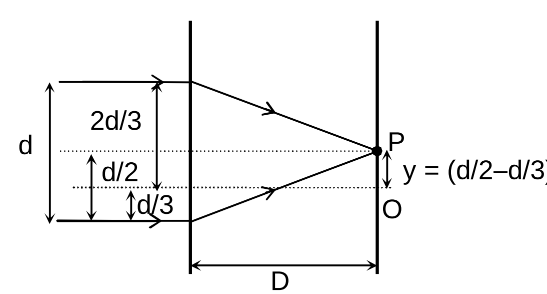
$\vec{v}_{BA} = \vec{v}_B - \vec{v}_A = \hat{i} - \left[-14(\hat{i}) + 10\hat{j}\right] = 15\hat{i} - 10\hat{j}$

$\left|\vec{v}_{BA}\right| = \sqrt{325}$ cm/sec

20. D

Sol. We know that P will be the central maxima (at which path difference is zero)

$OP = \frac{d}{2} - \frac{d}{3} = \frac{d}{6}$



SECTION – B

21. 00000.40

Sol. Error in measuring 50 sec = $\frac{1}{5} = 0.2\text{ sec}$

% error = $\frac{0.2}{50}\times 100 = 0.4\,\%$

22. 00000.00

Sol. In reverse biased diode offers infinite resistance

23. 00002.00

Sol. $L.C = \left(1-\frac{8}{10}\right) = 0.2\text{ mm} = 2\text{ cm}$

24. 00004.00

Sol. $\tan\theta = \frac{B_V}{B_H}$

25. 00000.21

Sol. $B_0 = \frac{E_0}{C}$

26. 00003.00

Sol. $\frac{q^2}{a^2} \propto Ta$

$a = k\left(\frac{q^2}{T}\right)^{1/3}$

27. 00003.00

Sol. N = 3

$W = \int PdV = \int nRkVdV$ …(i)

$T = kV^2 \Rightarrow dT = 2kVdV$ …(ii)

From (i) and (ii)

$W = nRk\frac{dT}{2k} = \frac{nR}{2}\int_{T_0}^{4T_0} dT = \frac{3}{2}nRT_0$

28. 00000.00

Sol. No change in flux.

29. 00003.00

Sol. Average value in one time period = $\frac{\int_0^T idt}{\int_0^T dt} = 3$

30. 00016.00

Sol. At surface $V = \frac{kQ}{R}$

at r < R, $V = \frac{kQ}{2R^3}(3R^2 - r^2)$

for r = R/2

$V = \frac{11}{8}\frac{kQ}{R}$

$\therefore\ \frac{1}{2}mv_0^2 = q\left[\frac{11\,kQ}{8R} - \frac{kQ}{R}\right]$

Chemistry

PART – B

SECTION – A

31. B
Sol. Ideal gaseous particles have no inter atomic interactions, so, zero potential energy.

32. C
Sol. $O_2F_2 \longrightarrow O_2^{2+}$ Bond order $= 3$
$KO_2 \longrightarrow O_2^-$ Bond order $= 1.5$
O_3 Bond order $= 1.5$
O_2 Bond order $= 2$
Higher bond order means lower bond length.

33. C
Sol. Ionization enthalpies is in the order : Li < B < Be < C < O < N < F < Ne.

34. C
Sol. $T_C = \frac{8a}{27Rb} \Rightarrow T_C \propto a$, and $Z = 1 - \left(\frac{a}{RT}\right)\frac{1}{V_m}$

35. A
Sol. Lithium is not miscible with other alkali metals.

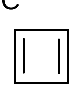
36. D
Sol. Beacause K has lower melting than Li, it melts easily and provide greater surface area during reaction with water.

37. A
Sol. $W = 0, \Delta U = q \Rightarrow nC_V\Delta T = q\left(C_V = \frac{3R}{2}\right)$

38. C
Sol. The pre-exponential factor does not increase with decrease in temperature.

39. B
Sol. Due to restriction on light and stopping of photosynthesis, plants start using oxygen which consequently decreases the oxygen inside the water.

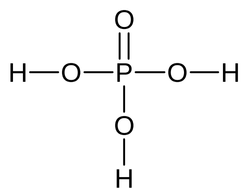
40. B
Sol. Increasing the concentrtion of conjugate base (or salt) decrease the H^+ concentration, thus increases pH.

41. C
Sol.  $\longrightarrow$ If two types of C – C bond length; it means there is no conjugation in this compound.

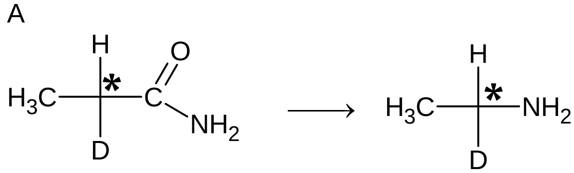
42. A
Sol. Paramagnetic compound :
$Na_3[CoF_6], CsO_2, [V(H_2O)_6]_3[FeF_6]_2, K_3[Fe(H_2O)_6]$

43. B
Sol. The slope change in the Ellingham diagram is due to phase change of metal. And, the higher values of ΔG^o generally show its inability to form oxide.

44. D
Sol. Zn has a less enthalpy of atomization because of low inter-atomic attraction.

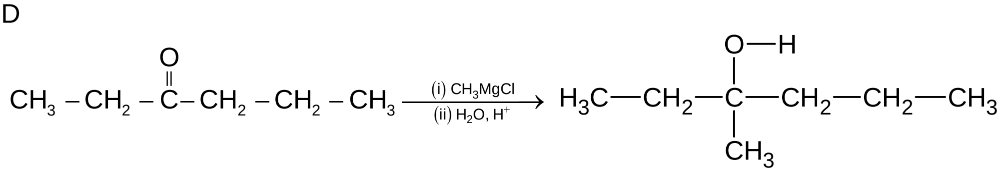
45. A
Sol.


46. C
Sol. During the given de-esterification reaction, the C – O bond of alcohol (here it is methanol) does not break. So, oxygen will remain O^{16} isotope.

47. A
Sol.


The configuration around the chiral carbon remain same.

48. A
Sol.
$$CH_3-\overset{\overset{\large O}{||}}{C}-CH_3 + 2Zn + 4HCl \longrightarrow CH_3-CH_2-CH_3 + 2ZnCl_2 + H_2O$$
(Balanced equation)

49. D
Sol.


50. B
Sol. In the given Newmann projection, after the rotating the group of atoms around C_2-C_3 bond, it will have plane of symmetry; so, it is not chiral.

SECTION – B

51. 00008.84

Sol. Let the mass be x gm

$$\frac{x}{180} \times \frac{1000}{(500 - x)} = 0.1$$

$$x = 8.84$$

52. 00011.40

Sol. Let the transition occurs between n_1 and n_2, ($n_2 > n_1$)
Number of waves in 2nd orbit of H-atom = 2
Degeneracy of 1st excited state of H-atom = 4

$$n_1 + n_2 = 4$$

$$n_2 - n_1 = 2$$

$$n_1 = 1, n_2 = 3$$

Find λ

53. 00099.65

Sol. Let $CH_4 \rightarrow$ x lit, then $C_2H_6 \rightarrow (20 - x)L$

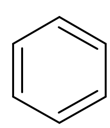
$$800 = \frac{x}{22.4} \times 894 + \frac{(20 - x)}{22.4} \times 1500$$

$$x = 19.93$$

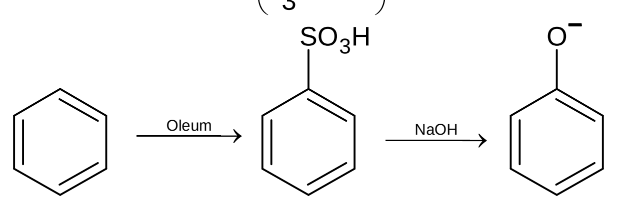
54. 00003.10

Sol. $$CaC_2 + 2H_2O \xrightarrow{50\%} Ca(OH)_2 + C_2H_2$$

(0.2 mole) (0.1 mole)

$3C_2H_2 \xrightarrow[873\text{ K}]{\text{Red hot iron}}$ 

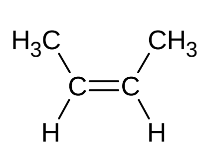
(0.1 mole) $\left(\frac{0.1}{3}\text{ mole}\right)$



$\left(\frac{0.1}{3}\text{ mole}\right)$

Required mass $= \left(\frac{0.1}{3} \times 93\right)\text{gm} = 3.1$

55. 00108.00

Sol. $H_3C—C\equiv C—CH_3 \xrightarrow{\text{Lindlar's catalyst}}$ 

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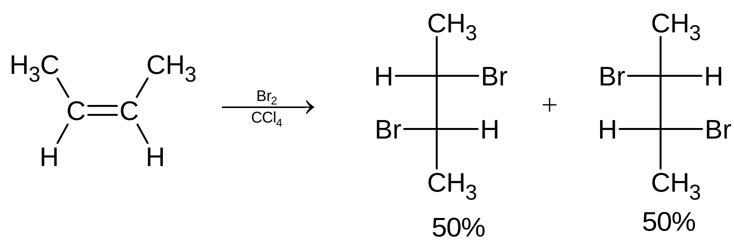


56. 00001.00

Sol. $\left(\Delta T_b\right)_A = K_b\left(A\right)\times m$

$\left(\Delta T_b\right)_B = K_b\left(B\right)\times m'$

$\Rightarrow m = 2m'$

And, $\left(\Delta T_b\right)_A = 2\left(\Delta T_b\right)_B$

Hence, $\dfrac{K_b\left(A\right)}{K_b\left(B\right)} = 1$

57. 00004.36

Sol. 422.63 gm of ZnS means 4.36 moles of ZnS.
There are 4 moles of ZnS in each mole of unit cell (in Zinc blende structure). Each unit cell also contain 4 unoccupied tetrahedral voids.

58. 00001.07

Sol. $E_{cell} = E^{o}_{cell} - \dfrac{0.059}{2}\log\dfrac{\left[Zn^{2+}\right]}{\left[Cu^{2+}\right]}$

$\Rightarrow \log\dfrac{\left[Zn^{2+}\right]}{\left[Cu^{2+}\right]} = \dfrac{2E^{o}}{0.059} - \dfrac{2E}{0.059}$

$\Rightarrow \left(y = C - mx\right)$

$\Rightarrow 1 = 37.28 - \dfrac{2E}{0.059}$

$\Rightarrow E = 1.07$

59. 00008.13

Sol. Passing through H_2S solution produces PbS and Ag_2S as precipitates, but PbS gets dissolved upon heating. Hence, only Ag_2S will remain as precipitate.

6.2 of $Ag_2S = \dfrac{0.1}{4}$ moles of Ag_2S

$= \dfrac{0.1}{4}$ mole of PbS

60. 00000.20

Sol. $\underset{\left(\text{dilute}\right)}{3Cu + 8HNO_3} \longrightarrow 3Cu\left(NO_3\right)_2 + 2NO + 4H_2O$

Mathematics PART – C

SECTION – A

61. B

Sol. $A + B = 2I$ ..... (1)

$A^{-1} + B^{-1} = 3I$ ..... (2)

Multiplying by matrix A in equation (2)

$AA^{-1} + AB^{-1} = 3AI \Rightarrow I + AB^{-1} = 3A$

Now multiplying by matrix B

$IB + AB^{-1}B = 3AB \Rightarrow B + A = 3AB$

$\Rightarrow 2I = 3AB \Rightarrow AB = \frac{2I}{3}$

62. C

Sol. $$\lim_{x\to 0} \frac{a\left(x - \frac{x^3}{3!} + \frac{x^5}{5!} - .....\right) - bx + cx^2 + x^3}{2x^2\left(x - \frac{x^2}{2} + \frac{x^3}{3} - .....\right) - 2x^3 + x^4} = \lim_{x\to 0} \frac{(a-b)x + cx^2 + \left(1 - \frac{a}{3!}\right)x^3 + \frac{a}{5!}x^5 - .....}{\frac{2x^5}{3} - \frac{2x^6}{4} + ...}$$

Limit is finite, if c = 0, a – b = 0, $1 - \frac{a}{3!} = 0$

63. D

Sol. $$\left|A^{2015} - 5A^{2014}\right| = \left|A^{2014}(A - 5I)\right| = \left|A^{2014}\right|\left|A - 5I\right| = (-1)^{2014}\begin{vmatrix} 0 & 2016 \\ 1 & 398 \end{vmatrix} = -2016$$

64. A

Sol. $$|\vec{a}\times\vec{b}| = \frac{|\vec{a}-\vec{b}|}{2} \Rightarrow |\vec{a}|^2|\vec{b}|^2 \sin^2\theta = \frac{1}{4}\left(|\vec{a}|^2 + |\vec{b}|^2 - 2(\vec{a}\cdot\vec{b})\right)$$

$$\Rightarrow \sin^2\theta = \frac{1}{4}(2 - 2\cos\theta) \Rightarrow 2\sin^2\theta = 1 - \cos\theta \Rightarrow \cos\theta = -\frac{1}{2}$$

$$\left|\vec{a}\times(\vec{a}\times\vec{b})\right|^2 = |\vec{a}|^2|\vec{a}\times\vec{b}|^2 = |\vec{a}|^2|\vec{b}|^2\sin^2\theta = \frac{3}{4}$$

65. C

Sol. $$I = \frac{1}{48}\int\left(48x^{23} + 48x^{15} + 48x^7\right)\left(2x^{24} + 3x^{16} + 6x^8\right)^{\frac{1}{8}}dx$$

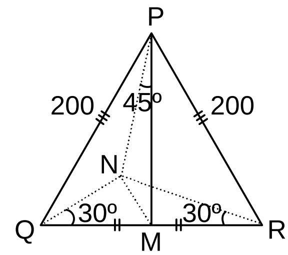
Let $2x^{24} + 3x^{16} + 6x^8 = y$

$$I = \frac{1}{48}\int(y)^{\frac{1}{8}}dy = \frac{1}{48}\cdot\frac{8}{9}(y)^{\frac{9}{8}} + c$$

$$I = \frac{1}{54}(y)^{\frac{9}{8}} + c$$

66. A

Sol. Let height of tower MN is h



In ΔQMN, $\tan 30^\circ = \dfrac{MN}{QM}$

$\Rightarrow \dfrac{1}{\sqrt{3}} = \dfrac{h}{QM}$

$\Rightarrow QM = \sqrt{3}h$

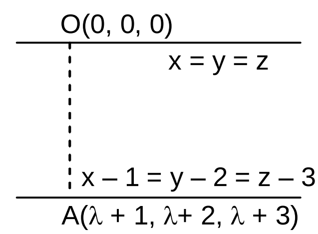
In ΔMNP, MN = PM

In ΔPMQ, $MP = \sqrt{(200)^2 - (\sqrt{3}h)^2} \Rightarrow h = \sqrt{40000 - 3h^2} \Rightarrow h^2 + 3h^2 = 40000$

$\Rightarrow 4h^2 = 40000 \Rightarrow h^2 = 10000 \Rightarrow h = 100$ meters

67. D

Sol. OA is perpendicular to line $L_2 = 0$



$1(\lambda + 1) + 1(\lambda + 2) + 1(\lambda + 3) = 0$

$\Rightarrow \lambda = -2$

A = (–1, 0, 1)

$\overrightarrow{OA} = -\hat{i} + \hat{k}$

$\because$ A normal vector to required plane is $\overrightarrow{OA}$

Equation of required plane is $-1(x - 0) + 1(z - 0) = 0$

$\Rightarrow -x + z = 0 \Rightarrow -10x + 10z = 0 \Rightarrow 10x - 10z = 0$

b = 0, c = –10, d = 0

68. B

Sol. $y = \tan^{-1}\left(\dfrac{5x - x}{1 + 5x(x)}\right) + \tan^{-1}\left(\dfrac{x + \frac{2}{3}}{1 - \frac{2x}{3}}\right) = \left(\tan^{-1}(5x) - \tan^{-1}(x)\right) + \left(\tan^{-1}(x) + \tan^{-1}\left(\dfrac{2}{3}\right)\right)$

$y' = \dfrac{5}{1 + 25x^2}$

69. A

Sol. (a, b) and (b, a) lie on circle

70. C

Sol. Ram does not get a first class, then he does not work hard

71. B

Sol. z = i or –i

72. D

Sol. $(1 - x)^5(1 + x)^4(1 + x^2)^4 = (1 - x)(1 - x^4)^4$

Coefficient of $x^{13} = (-1)(-4) = 4$

73. B

Sol. $4\cos 36^\circ = 4\left(\dfrac{\sqrt{5} + 1}{4}\right) = \sqrt{5} + \sqrt{1}$

$\cot\left(7\dfrac{1}{2}^\circ\right) = \dfrac{1 + \cos(15^\circ)}{\sin 15^\circ} = \dfrac{1 + \left(\dfrac{\sqrt{3} + 1}{2\sqrt{2}}\right)}{\dfrac{\sqrt{3} - 1}{2\sqrt{2}}} = \sqrt{6} + \sqrt{3} + \sqrt{2} + \sqrt{4}$

$$\sum_{i=1}^{6}\left(n_i\right)^2 = 1^2+2^2+3^2+4^2+5^2+6^2 = \frac{1}{6}\times 6\times 7\times 13 = 91$$

74. A

Sol. $T_n = \tan^{-1}\left[\frac{n^3-(n-1)^3}{1+n^3(n-1)^3}\right]$

$T_n = \tan^{-1}(n^3) - \tan^{-1}((n-1)^3)$

75. B

Sol. Arithmetic Mean = $\frac{0.1+1\cdot {}^nC_1+2\cdot {}^nC_2+.....+n\cdot {}^nC_n}{1+{}^nC_1+{}^nC_2+.....+{}^nC_n} = \frac{n\cdot(2)^{n-1}}{2^n} = \frac{n}{2}$

76. A

Sol. $f(\theta) = (\cos\theta + 1) - (\cos\theta - 1) - (\cos\theta - 2) - (\cos\theta - 3) = 7 - 2\cos\theta$

$f(\theta)_{max} = 9$; $f(\theta)_{min} = 5$

77. D

Sol. $\lambda^2 - 4\lambda + 3 = 0 \Rightarrow \lambda = 1, 3$

$\lambda^2 - 5\lambda + 6 = 0 \Rightarrow \lambda = 2, 3$

$\lambda^2 - 9 = 0 \Rightarrow \lambda = \pm 3$; $\lambda = 3$

78. D

Sol. $x = \sec\theta$; $y = \operatorname{cosec}\theta$

$\therefore \frac{5x+12y+7xy}{xy} = \frac{5}{y}+\frac{12}{x}+7 = 5\sin\theta + 12\cos\theta + 7$

79. B

Sol. $\frac{p}{q} = \sum_{n=1}^{360}\left(\frac{1}{\sqrt{n}\sqrt{n+1}\left(\sqrt{n}+\sqrt{n+1}\right)}\right) = \sum_{n=1}^{360}\frac{\sqrt{n+1}-\sqrt{n}}{\sqrt{n}\sqrt{n+1}(n+1-n)} = \sum_{n=1}^{360}\left(\frac{1}{\sqrt{n}}-\frac{1}{\sqrt{n+1}}\right)$

$= \left(1-\frac{1}{\sqrt{2}}\right)+\left(\frac{1}{\sqrt{2}}-\frac{1}{\sqrt{3}}\right)+....+\left(\frac{1}{\sqrt{360}}-\frac{1}{\sqrt{360}}\right) = 1-\frac{1}{19} = \frac{18}{19}$

$\therefore |p - q| = 1$

80. C

Sol. Let $g(x) = f^{-1}(x) = x^3 + 3x - 4$

$\Rightarrow f(g(x)) = x \Rightarrow f'\left(g(x)\right) = \frac{1}{g'(x)}$

$f''\left(g(1)\right) = -\frac{g''(1)}{\left(g'(1)\right)^3} \Rightarrow f''(0) = -\frac{6}{(6)^3} = -\frac{1}{36}$

$\Rightarrow \left|36f''(0)\right| = 1$

SECTION – B

81. 00004.50

Sol. $I = \int_1^4 \frac{-x^2\left(f'(x)+g'(x)\right)}{x^2}dx = -\left[f(x)+g(x)\right]_1^4 = -\left(f(4)+g(4)-9e\right)$

Now, $f(x) + g(x) = -x^2(f'(x) + g('(x))$

$\Rightarrow \int\frac{f'(x)+g'(x)}{f(x)+g(x)}dx = -\int x^{-2}dx = \frac{1}{x}+c$

$\Rightarrow$ f(x) + g(x) = $ae^{\frac{1}{x}}$ $\Rightarrow$ a = 9

$I = -\left(9e^{\frac{1}{4}} - 9e\right) = 9\left(e - e^{\frac{1}{4}}\right)$

82. 00001.00

Sol. $y' = \left\{-a\sin\left(e^x\right) + b\cos\left(e^x\right)\right\}e^x$

$y'' = \left\{-a\cos\left(e^x\right) - b\sin\left(e^x\right)\right\}e^x \cdot e^x + \left\{-a\sin\left(e^x\right) + b\cos\left(e^x\right)\right\}e^x = -ye^{2x} + y'$

$\Rightarrow \dfrac{y'' + ye^{2x}}{y'} = 1$

83. 00008.00

Sol. $a_1e_1 = a_2e_2$

$PF_1^2 + PF_2^2 = 4a_1^2e_1^2$

$PF_1 + PF_2 = 2a_1$

$\left|PF_1 - PF_2\right| = 2a_2$

$\left(PF_1 + PF_2\right)^2 + \left(PF_1 - PF_2\right)^2 = 4\left(a_1^2 + a_2^2\right)$

$\Rightarrow 2\left(PF_1^2 + PF_2^2\right) = 4\left(a_1^2 + a_2^2\right) \Rightarrow 4a_1^2e_1^2 = 2\left(a_1^2 + a_2^2\right)$

$\Rightarrow 2e_1^2 = 1 + \left(\dfrac{a_2}{a_1}\right)^2 \Rightarrow 2e_1^2 = 1 + \left(\dfrac{e_1}{e_2}\right)^2 \Rightarrow \dfrac{1}{e_1^2} + \dfrac{1}{e_2^2} = 2$

Let $\dfrac{1}{e_1} = \sqrt{2}\cos\theta$; $\dfrac{1}{e_2} = \sqrt{2}\sin\theta$

$E = 9e_1^2 + e_2^2 = \dfrac{9}{2}\sec^2\theta + \dfrac{1}{2}\text{cosec}^2\theta$

$E = \dfrac{1}{2}\left(10 + 9\tan^2\theta + \cot^2\theta\right)$; $E_{min} = \dfrac{1}{2}(10 + 6) = 8$

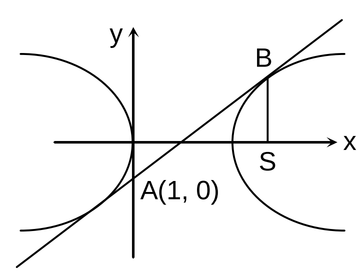
84. 00002.50

Sol. $\dfrac{x_1 + 2x_2 + 3x_3 + 4x_4 + 5x_5}{15} \geq \left(x_1 \cdot x_2^2 \cdot x_3^3 \cdot x_4^4 \cdot x_5^5\right)^{\frac{1}{15}}$

$\Rightarrow x_1 = x_2 = x_3 = x_4 = x_5 = 1$

85. 00028.00

Sol. Tangent to parabola is $y = mx - \dfrac{a}{4m}$ (1, 0)



$O = m - \dfrac{a}{4m} \Rightarrow a = 4m^2$

(1, 0) lies on the directrix of the hyperbola

$\dfrac{2}{e} = 1 \Rightarrow e = 2$

$\sqrt{4 + b^2} = 2 \times 2 \Rightarrow b^2 = 12$

Tangent to hyperbola $y = mx \pm \sqrt{4m^2 - b^2}$

$\therefore (-m)^2 = 4m^2 - b^2 \Rightarrow 3m^2 = b^2 = 12 \Rightarrow m^2 = 4$

$a + b^2 = 4m^2 + 12 = 28$

86. 00004.10

Sol. Let $P(E_1) = x$; $P(E_2) = y$; $P(E_3) = z$

$\therefore 3x(1 - y)(1 - z) = (1 - x)y(1 - z) = 9(1 - x)(1 - y)z = 3(1 - x)(1 - y)(1 - z)$

$$\Rightarrow \frac{3x}{1-x} = \frac{y}{1-y} = \frac{9z}{1-z} = 3 \Rightarrow x = \frac{1}{2}\ ;\ y = \frac{3}{4}\ ;\ z = \frac{1}{4}$$

$$\begin{vmatrix} \frac{1}{2} & \frac{3}{4} & \frac{1}{4} \\ \frac{3}{4} & \frac{1}{4} & \frac{1}{2} \\ \frac{1}{4} & \frac{1}{2} & \frac{3}{4} \end{vmatrix} = \frac{1}{64}\begin{vmatrix} 2 & 3 & 1 \\ 3 & 1 & 2 \\ 1 & 2 & 3 \end{vmatrix} = -\frac{9}{32}$$

87. 00002.00

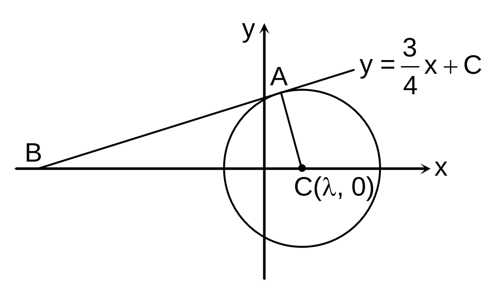
Sol. Clearly all the solutions of f(x) = x are also solutions of f(f(x)) = x. First, we solve f(x) = x, f(x) = 0
$\Rightarrow x^2 - 3x + 3 = x \Rightarrow x^2 - 4x + 3 = 0 \Rightarrow (x - 1)(x - 3) = 0 \Rightarrow x = 1, 3$
Therefore x = 1, 3 are also solutions of f(f(x)) = x. We want to seek if there are any more solutions of f(f(x)) = x other than 1 and 3.
$f(f(x)) = x \Rightarrow f(x^2 - 3x + 3) = x \Rightarrow (x^2 - 3x + 3)^2 - 3(x^2 - 3x + 3) + 3 = 0$
$\Rightarrow x^4 - 6x^3 + 12x^2 - 9x + 3 = 0 \Rightarrow (x^2 - 4x + 3)(x^2 - 2x + 1) = 0$
$\Rightarrow (x - 1)(x - 3)(x - 1)^2 = y \Rightarrow x = 1, 3$
In this case we have no additional solutions. Therefore the probability that x satisfies equation f(f(x)) = x is $\frac{2}{9}$, therefore m = 2

88. 00007.00

Sol. $(x^2 - 5x + 10)(x^2 - 5x + 2) + 12 = 0$
$\Rightarrow (x^2 - 5x + 10)(x^2 - 5x + 10 - 8) + 12 = 0$
$\Rightarrow (x^2 - 5x + 10)^2 - 8(x^2 - 5x + 10) + 12 = 0$
$x^2 - 5x + 10 = 2, 6$
Required sum = $\frac{1}{2} + \frac{1}{2} + \frac{1}{6} + \frac{1}{6} = \frac{4}{3}$

89. 00002.30

Sol. $$\left| \frac{\frac{3}{4}\lambda - 0 + C}{\sqrt{\left(\frac{3}{4}\right)^2 + (1)^2}} \right| = 5$$



$$\Rightarrow \left| \left(\frac{3\lambda + 4C}{4}\right) \times \frac{4}{5} \right| = 5$$

$$\Rightarrow \left| \frac{3\lambda + 4C}{3} \right| = \frac{25}{3}$$

$$|AB| = \sqrt{\left(\lambda + \frac{4C}{3}\right)^2 - 25} = \sqrt{\left(\frac{3\lambda + 4C}{3}\right)^2 - 25} = \sqrt{\left(\frac{25}{3}\right)^2 - 25} = \frac{20}{3}$$

90. 00002.80

Sol. $$\left(2 + x^3(3 + x)^2\right)^{10}$$
$$\downarrow$$
$$^{10}C_r \cdot (2)^{10-r} \cdot x^{3r}(3 + x)^{2r}$$
$$\downarrow$$
$$^{2r}C_k \cdot 3^{2r-k} \cdot x^k$$

$3r + k = 8 \Rightarrow r = k = 2$
$\therefore$ Coefficient of $x^8 = {}^{10}C_2 \cdot 2^8 \cdot {}^4C_2 \cdot 3^2 = 5 \cdot 2^9 \cdot 3^5$